Lecture 8. Numerical Integration

Newton–Cotes rules, Gaussian quadrature, orthogonal polynomials, Laplace approximation

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We want $\int_a^b f(x)\,dx$. Statistical integrals rarely have closed forms:

- expectations, $E\sin X = \int \sin(x) f_X(x)\,dx$;
- Bayesian normalizing constants $\int f(x \mid \theta)\pi(\theta)\,d\theta$;
- likelihoods of mixed models, e.g. a Poisson GLM with random intercept, $\prod_i \int \prod_j f(y_{ij} \mid \gamma_i)\phi(\gamma_i)\,d\gamma_i$.

Approaches: quadrature (now) and Monte Carlo (Lecture 10).

8.1 Newton–Cotes rules

With $h = (b-a)/n$, $x_i = a + ih$, approximate f on each $[x_i, x_{i+1}]$ by an interpolating polynomial of degree m, integrate it exactly, and add up.

- $m = 0$, **Riemann**: $\int_{x_i}^{x_{i+1}} f \approx h f(x_i)$.
- $m = 1$, **trapezoidal**: $\int_{x_i}^{x_{i+1}} f \approx \frac{h}{2}\{f(x_i) + f(x_{i+1})\}$, so $T(n) = h\{\frac{1}{2}f(a) + f(x_1) + \cdots + f(x_{n-1}) + \frac{1}{2}f(b)\}$.
- $m = 2$, **Simpson**: $\int_{x_i}^{x_{i+1}} f \approx \frac{h}{6}\{f(x_i) + 4f(\frac{x_i + x_{i+1}}{2}) + f(x_{i+1})\}$.

Proposition 8.1 (Accuracy)

For smooth f the total errors are $O(n^{-1})$ (Riemann), $O(n^{-2})$ (trapezoidal), $O(n^{-4})$ (Simpson); the local errors are

$$\frac{h^2}{2}f'(x_i) + O(h^3), \qquad -\frac{h^3}{12}f''(x_i) + O(h^4), \qquad -\frac{h^5}{2880}f^{(4)}(x_i) + O(h^6).$$

Proof. Integrate the Taylor expansion of f at x_i over $[x_i, x_i + h]$:

$$\int_{x_i}^{x_{i+1}} f = h f(x_i) + \frac{h^2}{2}f'(x_i) + \frac{h^3}{6}f''(x_i) + \frac{h^4}{24}f^{(3)}(x_i) + \frac{h^5}{120}f^{(4)}(x_i) + O(h^6).$$

Riemann: error $\frac{h^2}{2}f'(x_i) + O(h^3)$. *Trapezoid*: $\frac{h}{2}\{f(x_i) + f(x_{i+1})\} = h f(x_i) + \frac{h^2}{2}f' + \frac{h^3}{4}f'' + O(h^4)$, error $(\frac{1}{6} - \frac{1}{4})h^3 f'' + O(h^4) = -\frac{h^3}{12}f''(x_i) + O(h^4)$. *Simpson*: expanding at midpoint and endpoint, the rule matches through h^4 and differs by $(\frac{1}{120} - \frac{5}{576})h^5 f^{(4)} = -\frac{h^5}{2880}f^{(4)}$. Summing $n = (b-a)/h$ local errors gives global orders h, h^2, h^4. ∎

Simpson needs about twice the evaluations per interval, but wins by orders of magnitude at equal cost. For $\int_0^2 e^{-x}dx = 1 - e^{-2} \approx 0.864665$ and $n = 10$ the errors are 8.9×10^{-2}, 2.9×10^{-3}, 4.8×10^{-7}; log–log slopes against n are -1, -2, -4.

Example 8.1 (An infinite range)

Newton–Cotes rules need a finite interval. For $\int_1^\infty e^{-x}x^{-1/2}dx$ substitute $t = 1/x$:

$$\int_1^\infty e^{-x}x^{-1/2}dx = \int_0^1 e^{-1/t}t^{-3/2}\,dt.$$

At $t = 0$ the integrand is $0 \cdot \infty =$ `NaN` in floating point, so trapezoid and Simpson break there. But $e^{-1/t}t^{-3/2} \to 0$ with all derivatives as $t \downarrow 0$; defining it as 0 at $t = 0$ gives a smooth integrand, and the rules converge at their usual rates to $\sqrt{\pi}\,\mathrm{erfc}(1) = \Gamma(\frac{1}{2}, 1) \approx 0.278806$. (Or use an open rule: midpoint, Gaussian quadrature.)

General m: interpolation. For distinct nodes $x_0, \dots, x_m$, the **Lagrange polynomials** $L_j(x) = \prod_{k \neq j} \frac{x - x_k}{x_j - x_k}$ satisfy $L_j(x_k) = \delta_{jk}$, so $p = \sum_j y_j L_j$ interpolates (x_j, y_j).

Theorem 8.2 (Polynomial interpolation)

For $m + 1$ distinct points there is a unique polynomial of degree at most m through $(x_0, y_0), \dots, (x_m, y_m)$.

Proof. Existence: Lagrange form. Another q gives $p - q$ of degree $\leq m$ with $m + 1$ roots, so $p = q$. ■

Corollary 8.3

If f is a polynomial of degree at most m, then $\int_a^b f = \sum_j w_j f(x_j)$ with $w_j = \int_a^b L_j(x)\,dx$.

Proof. By uniqueness, $f = \sum_j f(x_j) L_j$; integrate. ■

8.2 Gaussian quadrature

Newton–Cotes fixes equally spaced nodes and chooses weights to integrate degree m exactly; choosing the nodes too gains much more.

$\mathcal{P}^m$, the polynomials of degree $\leq m$ on (a, b), has inner product

$$\langle p, q \rangle = \int_a^b p(x) q(x) w(x)\,dx$$

for a positive weight w. Gram–Schmidt on $1, x, x^2, \dots$ gives orthonormal $q_0, q_1, \dots, q_k$ of exact degree k and orthogonal to $\mathcal{P}^{k-1}$; for $w \equiv 1$ on $(-1, 1)$, the **Legendre polynomials**.

Lemma 8.4

The roots of q_k are real, simple and lie in (a, b).

Proof. Let $r_1 < \cdots < r_l$ be the sign changes of q_k in (a, b). If $l < k$, $q_k \prod_{i \leq l} (x - r_i)$ has constant sign and is not 0, so its integral is nonzero, yet $\prod_{i \leq l} (x - r_i) \in \mathcal{P}^{k-1}$ is orthogonal to q_k. So $l = k$: k distinct real roots in (a, b). ■

Theorem 8.5 (Gaussian quadrature)

Let $x_0, \dots, x_m$ be the roots of q_{m+1} and $w_j = \int_a^b L_j(x) w(x)\,dx$. Then

$$\int_a^b f(x) w(x)\,dx = \sum_{j=0}^{m} w_j f(x_j)$$

for every polynomial f of degree at most $2m + 1$, and all $w_j > 0$. Degree $2m + 1$ cannot be improved.

Proof. Divide: $f = g q_{m+1} + r$, $g, r \in \mathcal{P}^m$. Then $\int g q_{m+1} w = 0$, so $\int f w = \int r w$, and $f(x_j) = r(x_j)$; by Corollary 8.3 (weight w) $\int r w = \sum w_j r(x_j) = \sum w_j f(x_j)$. Positivity: L_k^2 has degree $2m$, so $0 < \int L_k^2 w = \sum_j w_j L_k(x_j)^2 = w_k$. Optimality: $f = \prod_j (x - x_j)^2$ has degree $2m + 2$ and $\int f w > 0$, while the rule gives 0. ■

$m + 1$ well-chosen nodes are exact to degree $2m + 1$, against m for equal spacing.

Weight functions. Other weights give other orthogonal families, for integrals $\int g(x)w(x)\,dx$:

polynomials	weight $w(x)$	domain	related distribution
Legendre	1	$(-1,1)$	uniform
Laguerre	$x^\alpha e^{-x}$	$(0,\infty)$	gamma
Hermite	e^{-x^2}	$(-\infty,\infty)$	normal
Jacobi	$(1-x)^\alpha(1+x)^\beta$	$(-1,1)$	beta (after rescaling)

Each satisfies a three-term recursion $q_m(x) = (\alpha_m + \beta_m x)q_{m-1}(x) - \gamma_m q_{m-2}(x)$ and has *interlacing* roots (those of q_{m+1} separate those of q_m); the nodes are found by Newton's method using these, or as eigenvalues of a tridiagonal matrix.

Gauss–Hermite quadrature. $\int f(x)\,dx = \int\{f(x)e^{x^2}\}e^{-x^2}dx \approx \sum_j w_j f(x_j)e^{x_j^2}$, good only if $f(x)e^{x^2}$ is nearly polynomial. For $f(x) = e^{-(x-\mu)^2/(2\sigma^2)}$ with μ far from 0 or σ^2 far from $\frac{1}{2}$, the nodes miss the mass. Remedy: *centre and scale*, $z = (x-\mu)/(\sqrt{2}\sigma)$ gives $\sqrt{2}\sigma\int e^{-z^2}dz$, exact with one node.

Example 8.2 (Binomial–logit posterior)

$x \sim B(n,p)$, $p = e^\theta/(1+e^\theta)$, prior $\theta \sim N(0,\sigma^2)$. The normalizing constant is $\int \bar{g}$, $\bar{g}(\theta) = e^{\theta x}(1+e^\theta)^{-n}e^{-\theta^2/(2\sigma^2)}$. Gauss–Hermite rescaled only by σ does badly for $x/n = 0.9$: the integrand peaks near $\theta = \log 9 = 2.2$, far from the nodes. Centring at the maximizer θ^*, $\int \bar{g} = \int\{\bar{g}(z+\theta^*)e^{z^2}\}e^{-z^2}dz$, is accurate with few nodes.

Example 8.3 (Poisson posterior: centre *and* scale)

$y_1,\ldots,y_{10}$ Poisson with mean e^α, prior $\alpha \sim N(0,100)$, data $11,19,27,12,14,11,10,13,15,10$ ($\sum y_i = 142$). The log unnormalized posterior $h(\alpha) = 142\alpha - 10e^\alpha - \alpha^2/200$ has mode $\hat{\alpha} = 2.65306$ and scale $\hat{s} = \{-h''(\hat{\alpha})\}^{-1/2} = 0.08392$. With $\alpha = \hat{\alpha} + \sqrt{2}\hat{s}z$,

$$E[g(\alpha)\mid y] = \frac{\int g(\hat{\alpha}+\sqrt{2}\hat{s}z)e^{h(\hat{\alpha}+\sqrt{2}\hat{s}z)-h(\hat{\alpha})+z^2}e^{-z^2}dz}{\int e^{h(\hat{\alpha}+\sqrt{2}\hat{s}z)-h(\hat{\alpha})+z^2}e^{-z^2}dz},$$

both by Gauss–Hermite ($\sqrt{2}\hat{s}$ cancels; subtracting $h(\hat{\alpha}) \approx 235$ prevents overflow). The posterior is nearly normal, so the bracketed functions are nearly polynomial:

nodes	posterior mean	posterior variance
2	2.651881	0.0070417
5	2.649534	0.0070677
10	2.6495297	0.00706793
15	2.649529692	0.007067929

Fifteen nodes match adaptive integration to 10 digits. Uncentred, the nodes sit near 0, where the posterior (mean 2.65, sd 0.084) is numerically zero, and the rule fails completely.

Example 8.4 (Variance of the sample median)

For $X_1,\ldots,X_n$ i.i.d. $N(0,1)$, $n = 2m+1$, the median $X_{(m+1)}$ has density $n\binom{n-1}{m}\Phi(x)^m\{1-\Phi(x)\}^m\phi(x)$ (one observation in $[x, x+dx)$, exactly m of the other $n-1$ below). By symmetry

$EX_{(m+1)} = 0$ and

$$\begin{aligned}\operatorname{Var} X_{(m+1)} &= \int_{-\infty}^{\infty} x^2 n \binom{n-1}{m} \Phi(x)^m \{1-\Phi(x)\}^m \phi(x)\, dx \\ &= \frac{2n}{\sqrt{\pi}} \binom{n-1}{m} \int u^2 \Phi(\sqrt{2}u)^m \{1-\Phi(\sqrt{2}u)\}^m e^{-u^2} du,\end{aligned}$$

with $x = \sqrt{2}u$. Gauss–Hermite with k nodes:

k	2	4	8	16	32	64	exact	$\frac{\pi}{2n}$
$n = 11$	0.117	0.238	0.234	0.161	0.1379	0.13716	0.13716	0.1428
$n = 21$	0.007	0.057	0.127	0.123	0.0840	0.07349	0.07326	0.0748

Convergence is slow and erratic: $\Phi^m(1-\Phi)^m$ has width of order $n^{-1/2}$, while the nodes suit unit variance; rescaling by the asymptotic sd fixes this. The asymptotic variance $1/\{4n\phi(0)^2\} = \pi/(2n)$ is 4.1% ($n = 11$) and 2.1% ($n = 21$) too large.

Example 8.5 (Generalized Gauss–Laguerre)

With $x = u^2$, $\int_0^\infty \sqrt{x} e^{-x} dx = 2\int_0^\infty u^2 e^{-u^2} du = \frac{\sqrt{\pi}}{2}$ $(= \Gamma(\frac{3}{2}))$. For $\int_0^\infty f(x)\sqrt{x}e^{-x}dx$ use the weight $x^{1/2}e^{-x}$ (generalized Laguerre, $\alpha = \frac{1}{2}$): for $f \equiv 1$ one node $(w_0 = \Gamma(\frac{3}{2}))$ is exact. For $f = \log x$, exact value $\Gamma'(\frac{3}{2}) = \frac{\sqrt{\pi}}{2}(2 - \gamma - 2\log 2) \approx 0.032338$, two errors remain.

(i) *Method error*: $\log x$ is not polynomial and is singular at 0, so the error decays only algebraically in m.

(ii) *Floating-point error*: nodes and weights are computed approximately, the smallest node is near 0 where $\log x$ is large, the largest weights may underflow, and the small result (0.03) arises by cancellation, magnifying rounding errors.

8.3 Laplace approximation

For $\int f(x)e^{-ag(x)}dx$ with large a and g uniquely minimized at 0, the integrand concentrates near 0, and $g(x) \approx g(0) + \frac{1}{2}g''(0)x^2$ gives

$$\int f(x)e^{-ag(x)}dx \approx f(0)e^{-ag(0)}\sqrt{\frac{2\pi}{ag''(0)}}:$$

integration becomes optimization.

Theorem 8.6 (Laplace)

Suppose (1) for every $\delta > 0$ there is $\rho > 0$ with $g(x) - g(0) \geq \rho$ for $|x| \geq \delta$;
(2) g is twice continuously differentiable near 0 with $g'(0) = 0$, $g''(0) > 0$;
(3) f is continuous near 0 with $f(0) > 0$;
(4) $\int |f|e^{-ag} < \infty$ for $a \geq a_0$. Then

$$\lim_{a\to\infty} \frac{\int f(x)e^{-ag(x)}dx}{f(0)e^{-ag(0)}\sqrt{2\pi/(ag''(0))}} = 1.$$

Proof. Take $g(0) = 0$. Given $\epsilon \in (0, \frac{1}{2}g''(0))$, choose δ with, for $|x| < \delta$,

$$(1-\epsilon)f(0) \leq f(x) \leq (1+\epsilon)f(0), \qquad \tfrac{1}{2}(g''(0) - 2\epsilon)x^2 \leq g(x) \leq \tfrac{1}{2}(g''(0) + 2\epsilon)x^2.$$

Tails. For $|x| \geq \delta$, $g \geq \rho$, so for $a \geq a_0$,

$$\int_{|x|\geq\delta} |f|e^{-ag} = \int_{|x|\geq\delta} |f|e^{-a_0 g} e^{-(a-a_0)g} \leq e^{-(a-a_0)\rho} \int |f|e^{-a_0 g} = O(e^{-\rho a}).$$

Centre, upper bound. $\int_{|x|<\delta} fe^{-ag} \leq (1+\epsilon)f(0)\int e^{-\frac{a}{2}(g''(0)-2\epsilon)x^2}dx = (1+\epsilon)f(0)\sqrt{\frac{2\pi}{a(g''(0)-2\epsilon)}}$. *Centre, lower bound.* $\int_{|x|<\delta} fe^{-ag} \geq (1-\epsilon)f(0)\int_{|x|<\delta} e^{-\frac{a}{2}(g''(0)+2\epsilon)x^2}dx$, the Gaussian tail over $|x| \geq \delta$ being $O(e^{-\lambda a})$, $\lambda = \frac{1}{2}(g''(0)+2\epsilon)\delta^2$. Multiply by $\sqrt{a}$ and let $a \to \infty$:

$$(1-\epsilon)f(0)\sqrt{\frac{2\pi}{g''(0)+2\epsilon}} \leq \liminf \sqrt{a}\int fe^{-ag} \leq \limsup \sqrt{a}\int fe^{-ag} \leq (1+\epsilon)f(0)\sqrt{\frac{2\pi}{g''(0)-2\epsilon}}.$$

Let $\epsilon \to 0$. ■

Example 8.6 (Stirling's formula)

With $x = tz$, $\Gamma(t+1) = t^{t+1}\int_0^\infty e^{-t(z-\log z)}dz$; $g(z) = z - \log z$ has minimum $g(1) = 1$ with $g''(1) = 1$, so $\Gamma(t+1) \sim \sqrt{2\pi}\, t^{t+1/2}e^{-t}$.

Posterior expectations. With log-likelihood $\ell_n(\theta)$, log-prior $\pi(\theta)$ and $\tilde{h} = \log h > 0$,

$$E[h(\Theta) \mid x] = \frac{\int e^{\tilde{h}+\ell_n+\pi}d\theta}{\int e^{\ell_n+\pi}d\theta}.$$

Apply Laplace separately at the maximizers $\tilde{\theta}$ of $\tilde{h}+\ell_n+\pi$ and $\hat{\theta}$ of $\ell_n+\pi$ (posterior mode):

$$E[h(\Theta) \mid x] \approx \exp\{(\tilde{h}+\ell_n+\pi)(\tilde{\theta}) - (\ell_n+\pi)(\hat{\theta})\}\sqrt{\frac{(\ell_n+\pi)''(\hat{\theta})}{(\tilde{h}+\ell_n+\pi)''(\tilde{\theta})}}.$$

For large n the relative error is $O(n^{-2})$: the leading errors of numerator and denominator cancel.

Example 8.7 (Beta moments)

For $\Theta \sim \text{Beta}(\alpha,\beta)$, exactly $E\Theta^k = \frac{\alpha(\alpha+1)\cdots(\alpha+k-1)}{(\alpha+\beta)\cdots(\alpha+\beta+k-1)}$. With $g(\theta) = -(\alpha-1)\log\theta - (\beta-1)\log(1-\theta)$, minimized at $\hat{\theta} = \frac{\alpha-1}{\alpha+\beta-2}$,

$$\int_0^1 \theta^{\alpha-1}(1-\theta)^{\beta-1}d\theta \approx \Big(\frac{\alpha-1}{\alpha+\beta-2}\Big)^{\alpha-1}\Big(\frac{\beta-1}{\alpha+\beta-2}\Big)^{\beta-1}\sqrt{\frac{2\pi(\alpha-1)(\beta-1)}{(\alpha+\beta-2)^3}},$$

and likewise with $\alpha+k$ for the numerator. As $\alpha,\beta \to \infty$ with $\alpha/(\alpha+\beta) \to p$, both the moment and its approximation tend to p^k.

Unlike quadrature, whose node count grows exponentially with dimension, Laplace extends directly to $\mathbb{R}^d$: $\int fe^{-ag} \approx f(\hat{x})e^{-ag(\hat{x})}(2\pi/a)^{d/2}\det\{\nabla^2 g(\hat{x})\}^{-1/2}$.

8.4 Selected exercises

Motivation. Simpson as extrapolated trapezoid, the three-term recursion, and the two-point Gauss rule by hand.

Exercise 8.1 (Simpson's rule as an extrapolated trapezoid)

Let $T(n)$ be the composite trapezoidal rule and $S(n)$ the composite Simpson rule on n intervals of length $h = (b-a)/n$ (Simpson using the midpoints $m_i = x_i + h/2$).
(a) Show that $S(n) = \frac{1}{3}\{4T(2n) - T(n)\}$.
(b) Suppose $T(n) = I + ch^2 + O(h^4)$, where $I = \int_a^b f$ and c does not depend on h. Show that $\frac{1}{3}\{4T(2n) - T(n)\} = I + O(h^4)$, in agreement with Proposition 8.1.

Solution. [Supplement]

(a) $T(2n)$ uses the x_i and the midpoints m_i, spacing $h/2$:

$$T(n) = h\Big\{\tfrac{1}{2}f(a) + \sum_{i=1}^{n-1} f(x_i) + \tfrac{1}{2}f(b)\Big\},$$

$$T(2n) = \frac{h}{2}\Big\{\tfrac{1}{2}f(a) + \sum_{i=1}^{n-1} f(x_i) + \sum_{i=0}^{n-1} f(m_i) + \tfrac{1}{2}f(b)\Big\}.$$

Hence

$$4T(2n) - T(n) = h\Big\{\tfrac{1}{2}f(a) + \sum_{i=1}^{n-1} f(x_i) + \tfrac{1}{2}f(b)\Big\} + 2h\sum_{i=0}^{n-1} f(m_i),$$

and dividing by 3,

$$\frac{4T(2n) - T(n)}{3} = \frac{h}{6}\Big\{f(a) + 2\sum_{i=1}^{n-1} f(x_i) + f(b) + 4\sum_{i=0}^{n-1} f(m_i)\Big\} = \sum_{i=0}^{n-1} \frac{h}{6}\{f(x_i) + 4f(m_i) + f(x_{i+1})\},$$

which is $S(n)$ (each interior node lies in two intervals).

(b) $T(2n) = I + \frac{c}{4}h^2 + O(h^4)$, so $4T(2n) - T(n) = 3I + O(h^4)$. This is *Richardson extrapolation* (cancelling the leading error); repeating it gives Romberg integration. ■

Exercise 8.2 (The three-term recursion)

Let $\langle p, q\rangle = \int_a^b pqw$ with a positive weight w having finite moments, and let $p_0 = 1, p_1, p_2, \ldots$ be the *monic* orthogonal polynomials (p_k of degree k with leading coefficient 1, orthogonal to $\mathcal{P}^{k-1}$); set $p_{-1} = 0$.
(a) Show that for $k \geq 0$

$$p_{k+1}(x) = (x - a_k)p_k(x) - b_k p_{k-1}(x), \qquad a_k = \frac{\langle xp_k, p_k\rangle}{\langle p_k, p_k\rangle}, \quad b_k = \frac{\langle p_k, p_k\rangle}{\langle p_{k-1}, p_{k-1}\rangle} > 0$$

(b_0 is irrelevant since $p_{-1} = 0$).
(b) For the Legendre weight $w \equiv 1$ on $(-1, 1)$, compute p_1 and p_2.

Solution. [Supplement]

(a)

Step 1. $q = p_{k+1} - xp_k$ has degree $\leq k$ (both monic of degree $k+1$), so $q = \sum_{j\leq k} c_j p_j$, $c_j = \langle q, p_j\rangle/\langle p_j, p_j\rangle$.

Step 2. $\langle xp, r\rangle = \langle p, xr\rangle$, so for $j \leq k$, as $\langle p_{k+1}, p_j\rangle = 0$,

$$\langle q, p_j\rangle = -\langle xp_k, p_j\rangle = -\langle p_k, xp_j\rangle.$$

Step 3. For $j \leq k-2$, $xp_j \in \mathcal{P}^{k-1}$, so $c_j = 0$. For $j = k$, $c_k = -a_k$. For $j = k-1$, $xp_{k-1} = p_k + (\text{degree} \leq k-1)$, so $\langle p_k, xp_{k-1}\rangle = \langle p_k, p_k\rangle$ and $c_{k-1} = -b_k$.

Step 4. So $p_{k+1} = xp_k - a_k p_k - b_k p_{k-1}$, with $b_k > 0$ a ratio of squared norms. Normalizing $q_k = p_k/\|p_k\|$ gives the recursion quoted in the lecture.

(b) By symmetry $a_0 = \int_{-1}^{1} x\,dx/2 = 0$, so $p_1 = x$; $a_1 = \int_{-1}^{1} x^3 dx / \int_{-1}^{1} x^2 dx = 0$ and $b_1 = \frac{2/3}{2} = \frac{1}{3}$, so $p_2(x) = x^2 - \frac{1}{3}$. ■

Exercise 8.3 (Two-point Gauss–Legendre rule)

(a) Using Exercise 8.2(b), derive the Gaussian quadrature rule with two nodes for $\int_{-1}^{1} f(x)\,dx$. Verify directly that it integrates $1, x, x^2, x^3$ exactly but not x^4.
(b) Transfer the rule to an interval $[c - r, c + r]$.

Solution. [Supplement]

(a) By Theorem 8.5 with $m = 1$, the nodes are the roots $x_{0,1} = \mp 1/\sqrt{3}$ of p_2, and

$$w_0 = \int_{-1}^{1} \frac{x - 1/\sqrt{3}}{-2/\sqrt{3}}\,dx = \int_{-1}^{1} \Big(\frac{1}{2} - \frac{\sqrt{3}}{2}x\Big)dx = 1,$$

and $w_1 = 1$ by symmetry: $\int_{-1}^{1} f \approx f(-1/\sqrt{3}) + f(1/\sqrt{3})$. Check: 1 gives 2; x, x^3 give 0 (odd); x^2 gives $\frac{1}{3} + \frac{1}{3} = \frac{2}{3}$. For x^4 the rule gives $\frac{2}{9} \neq \frac{2}{5}$: exactness stops at $2m + 1 = 3$.

(b) With $x = c + rt$, $\int_{c-r}^{c+r} f\,dx = r \int_{-1}^{1} f(c + rt)\,dt \approx r\{f(c - r/\sqrt{3}) + f(c + r/\sqrt{3})\}$, exact for cubics: two evaluations, where Simpson needs three. ■